An Unofficial Compendium of Worked Solutions to the Problems in *Geometry*

by I. M. Gelfand & T. Alekseyevskaya (Gelfand)

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Abstract

This book is an independent and unofficial collection of worked solutions to problems inspired by the book *Geometry* by I. M. Gelfand and T. Alekseyevskaya (Gelfand), first published in 2020.

The original text presents a wide range of problems in projective geometry, affine geometry, symplectic geometry, and Euclidean geometry. The book contains solved exercises and questions, but the reader is also given 230 problems to solve.

The aim of this book is to provide clear, logically consistent solutions to all of the 230 unsolved problems addressed, written in a style accessible to motivated students and self-learners.

Figures are either drawn by hand and scanned, constructed using GeoGebra, or produced directly in LaTeX. Where applicable, proofs or calculations may also be checked using computer programs written in Scheme (a dialect of Lisp) or Python.

Large language models were used only as auxiliary tools, primarily for generating figures in LaTeX and for reviewing exposition.

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Issues and corrections may be reported at the project repository.

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Unofficial companion.

This document is an **independent, unofficial collection of worked solutions** to problems from *Geometry* by I. M. Gelfand and T. Alekseyevskaya (Gelfand) (first published in 2020).

This work is **not affiliated with, endorsed by, or approved by** the original authors or the publisher. The names of the authors and the title of the book are used only for identification and reference.

The original problem statements are not reproduced here. All exposition and solutions are written independently by the author of this document.

Project repository:

https://github.com/deepak-venkatesh/gelfand-geometry

For

My Daughter and Son

Contents

Working Notes

Challenging Problems for Middle Schoolers:

- Part I - Chapter 2 - Problem 7
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Typographical Errors:

1. The standard mathematical spelling of Hermann Minkowski's name is *Minkowski* and not *Minkowsky*.

Introduction

No problems in this chapter.

Part I

Points and Lines: A Look at Projective Geometry

Chapter 1. Points and lines

Problem 1.

In a line if we place a point we divide it into 2 parts. In this case both the part are *rays* and the point is an *endpoint*.

Now if there are 2 point on the line, we still have 2 rays but the two endpoints create a line segment.

So n points on a line will give 2 rays and $(n-1)$ line segments.

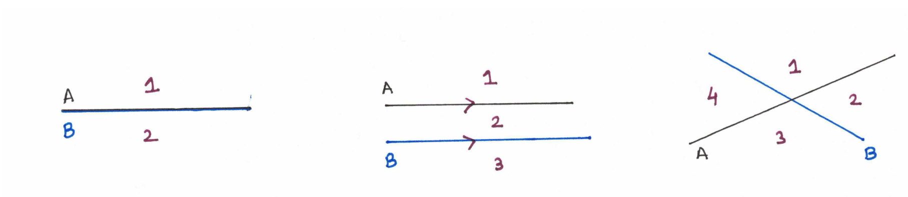
For this problem.

$$\begin{aligned}&n = 100\\&\text{endpoints} = 100\\&\text{rays} = 2\\&\text{segments} = 100 - 1 = 99\end{aligned}$$

Problem 2.

Three possibilities exist:
1. The two lines A and B can perfectly coincide.
2. They can be parallel to each other.
3. They can intersect at some point.

Basis the figure below the plane can be divided into 2, 3, or 4 parts.



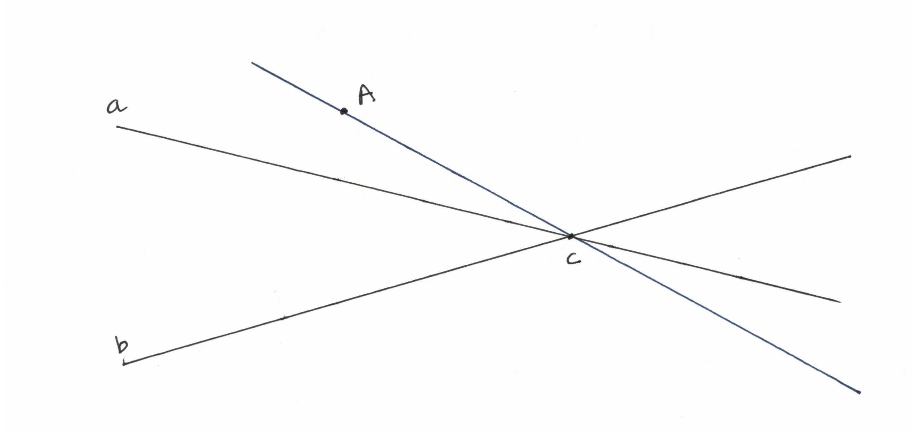
Problem 3.

The chapter explains that geometric constructions performed using the two axioms below are call constructions in *projective geometry*.

Axiom 1: Through any two points, one can draw a unique line.

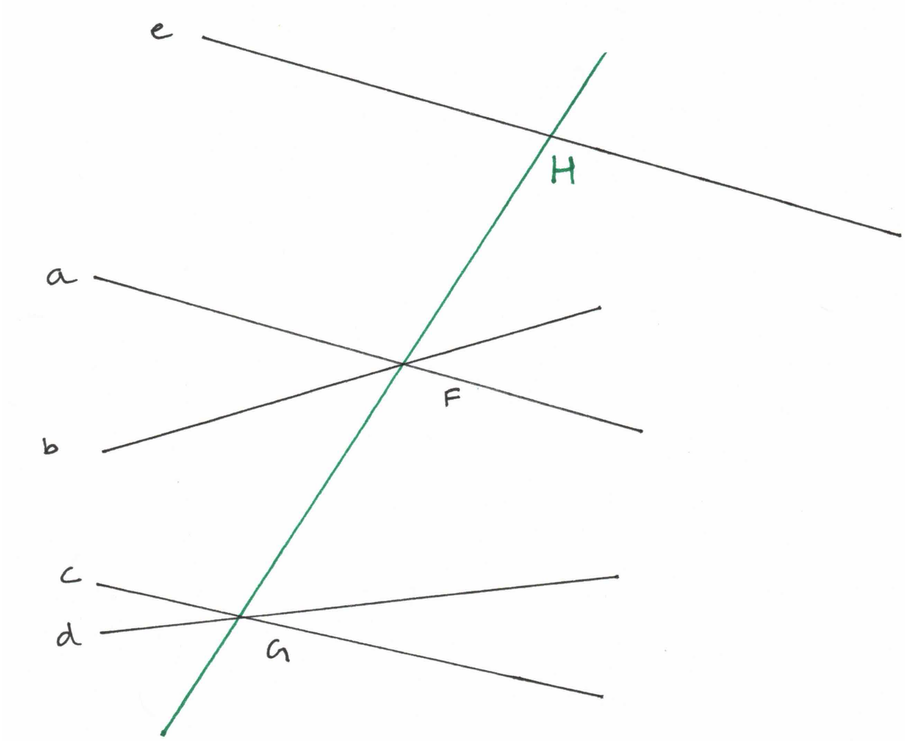
Axiom 2: Any two lines either have a unique point of intersection or do not intersect (are parallel).

Basis these two axioms of projective geometry we can construct the line AC.



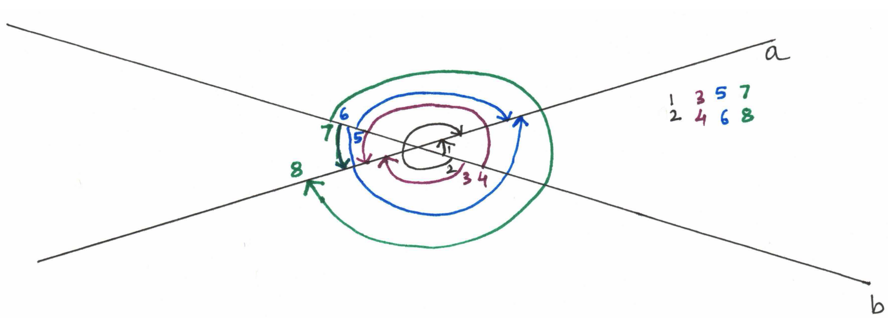
Problem 4.

F and G can be connected with a line since basis Axiom 1 of Projective Geometry we can construct a line. This line intersects at a unique point H with the line e. If e was parallel to line GF then they would never intersect. But given the visual in the problem we are assuming that they will intersect.



Problem 5.

There are 8 angles in this. From one side of the ray of b to one side of a (via convex) and another via not the convex side. Similarly 2 ways to other side ray a. Then 4 more ways for the other side of ray b.



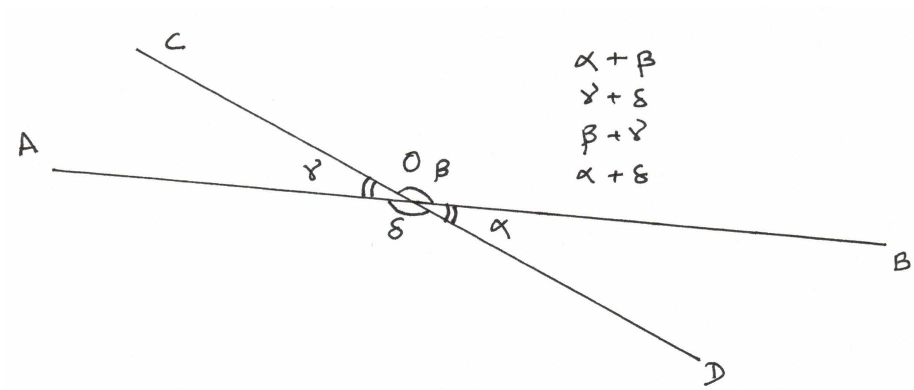
Problem 6.

The definition of supplementary angles is important. To quote the authors

> Two convex angles that have one side in common are positioned in such a way that the second side of one angle is an extension of the second side of the other angle are called *supplementary angles*.

In schools we are usually taught that if the sum of two angles is 180° they are called supplementary angles.

1. $\angle AOC$ and $\angle COB$
2. $\angle COB$ and $\angle BOD$
3. $\angle BOD$ and $\angle DOA$
4. $\angle DOA$ and $\angle AOC$

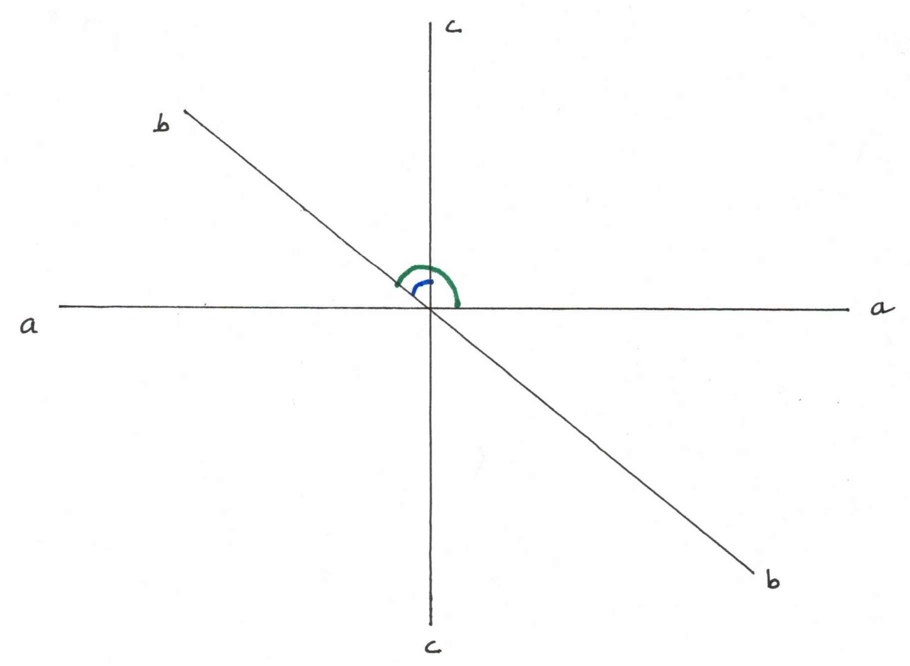


Problem 7.

This is a good problem. We get the following:
a) 6 straight angles (got by both sides of each line)
b) 6 adjacent convex angles
c) 6 convex angles formed between two rays skipping one in between

Any other skipping will form one of the three cases listed above. The figure shows the three lines.

Additionally, it is important that the lines intersect at a single point. If it did not they we would have overlapping lines and different answers to this question.


c
b
a
a
b
c

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Chapter 3. Three lines

Chapter 4. Four lines. Quadrilaterals

Chapter 5. Five lines

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Chapter 7. Dual configurations in projective geometry

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Chapter 9. Dual Desargues configuration

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